20260508_AKIB2_Weiterfuehrende_KI_Programmierung

Cart (Classification & Regression Trees)

ENTSCHEIDUNGSBÄUME

Day	outlook	temperature	humidity	wind	Decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

Verschmutzung der Information. misst die Homogenität eine Datasets.

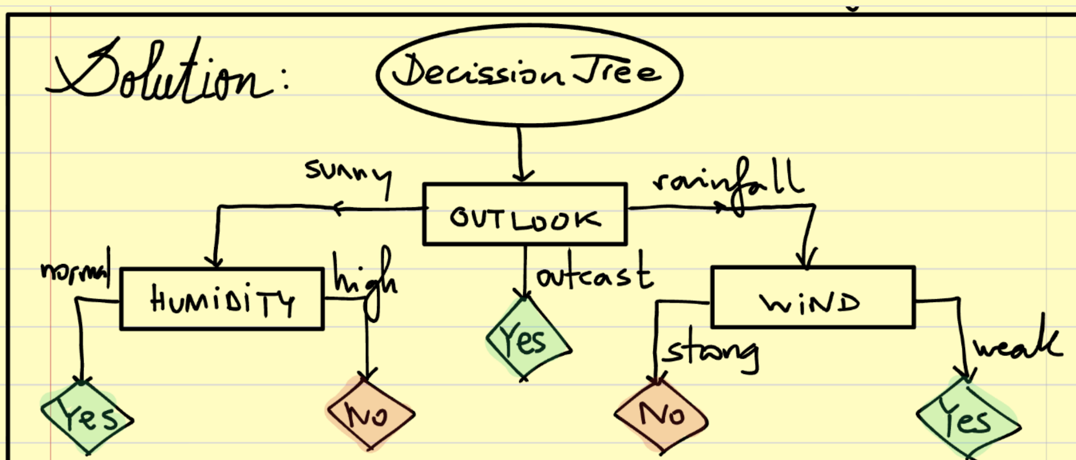
Die Informationsverschmutzung wird durch den GINI. Index gemessen:

$$\boxed{GINI = 1 - \sum_{i=1}^{n} p_i^2}$$

GINI = 0 die Daten sind pur

GINI = 1 die Daten sind schmutzig

Entscheidungsbaum:



Schritt 1. Wir schauen auf Allen Kriterien und nehmen den mit dem geringsten GINI.

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1	sunny	hot	high	weak	No
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3	overcast	hot	high	weak	Yes
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

1. OUTLOOK. sunny, overcast, rainfall

OUTLOOK	YES	No	#
sunny	2	3	5
overcast	4	0	4
rainfall	3	2	5
			14

$$\text{Gini}(\text{Outlook sunny}) = 1 - \sum p_i^2 = 1 - \left[\left(\overset{Y}{\frac{2}{5}}\right)^2 + \left(\frac{3}{5}\right)^2 \right] = 0'48$$

$$\text{Gini}(\text{Outlook overcast}) = 1 - \left[\left(\frac{4}{4}\right)^2 + \left(\frac{0}{4}\right)^2 \right] = 0$$

$$\text{Gini}(\text{Outlook rainfall}) = 1 - \left[\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 \right] = 0'48$$

$$\text{Gini}(\text{Outlook}) = \frac{5}{14} \cdot 0'48 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0'48 = 0'342$$

2. Temperature (hot, mild, cool)

	Yes	No	#
hot	2	2	4
mild	4	2	6
cool	3	1	4/14

$$\text{Gini}(\text{Temperature}) = \frac{4}{14} \cdot \overset{hot}{\left[1 - \left[\left(\overset{Y}{\frac{2}{4}}\right)^2 + \left(\overset{N}{\frac{2}{4}}\right)^2 \right] \right]} + \frac{6}{14} \cdot \overset{mild}{\left[1 - \left[\left(\frac{4}{6}\right)^2 + \left(\frac{2}{6}\right)^2 \right] \right]} + \frac{4}{14} \cdot \overset{cool}{\left[1 - \left[\left(\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^2 \right] \right]} = 0'439$$

3. Humidity . high/normal

	Yes	no	#
high	3	4	7
normal	6	1	7
			/14

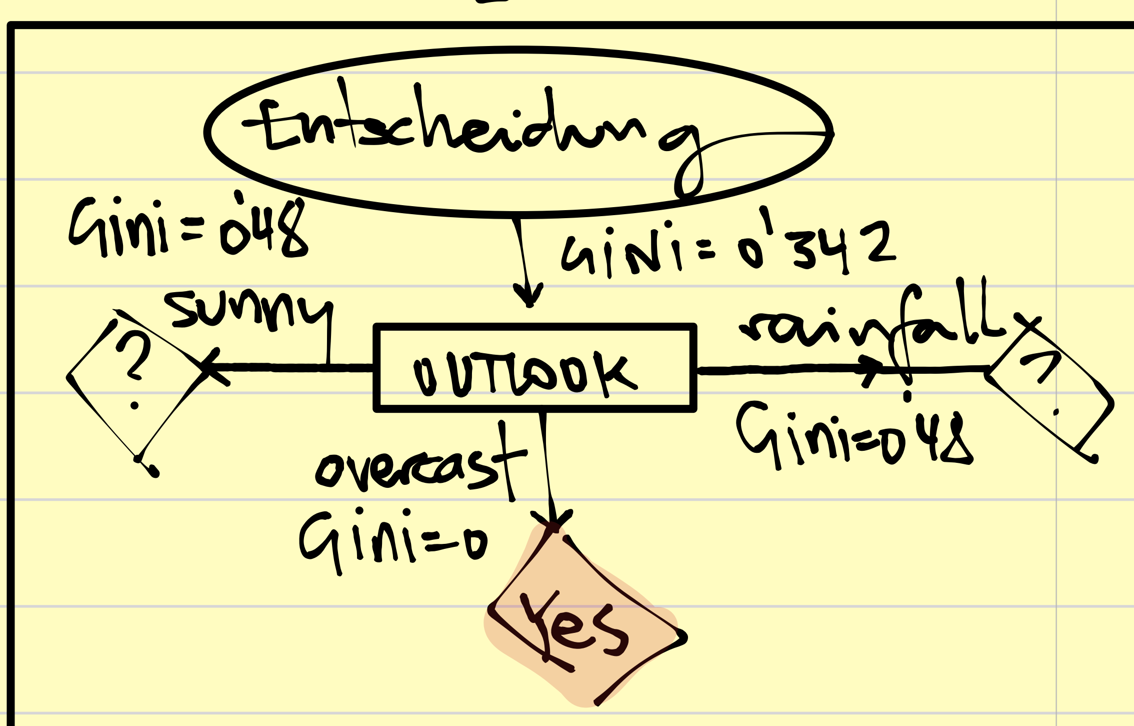
$$\text{Gini}(\text{Humidity}) = \frac{7}{14}\cdot\left[1-\left(\left(\frac{3}{7}\right)^2+\left(\frac{4}{7}\right)^2\right)\right]_{\text{high}} + \frac{7}{14}\cdot\left[1-\left(\left(\frac{6}{7}\right)^2+\left(\frac{1}{7}\right)^2\right)\right]_{\text{normal}} = 0'367$$

4. Wind . weak/strong

	Yes	No	#
weak	3	3	6
strong	6	2	8
			/14

$$\text{Gini}(\text{Wind}) = \frac{6}{14}\cdot\left[1-\left(\left(\frac{3}{6}\right)^2+\left(\frac{3}{6}\right)^2\right)\right]_{\text{weak}} + \frac{8}{14}\cdot\left[1-\left(\left(\frac{6}{8}\right)^2+\left(\frac{2}{8}\right)^2\right)\right]_{\text{strong}} = 0'428$$

	Gini
OUTLOOK	0'342
TEMP	0'439
HUM	0'367
WIND	0'428



Schritt 2. Wiederholung aus dem ersten Knote.

1. OUTLOOK SUNNY + Temp

	Yes	No	#
hot	0	2	2
mild	1	1	2
cool	1	0	1

hot

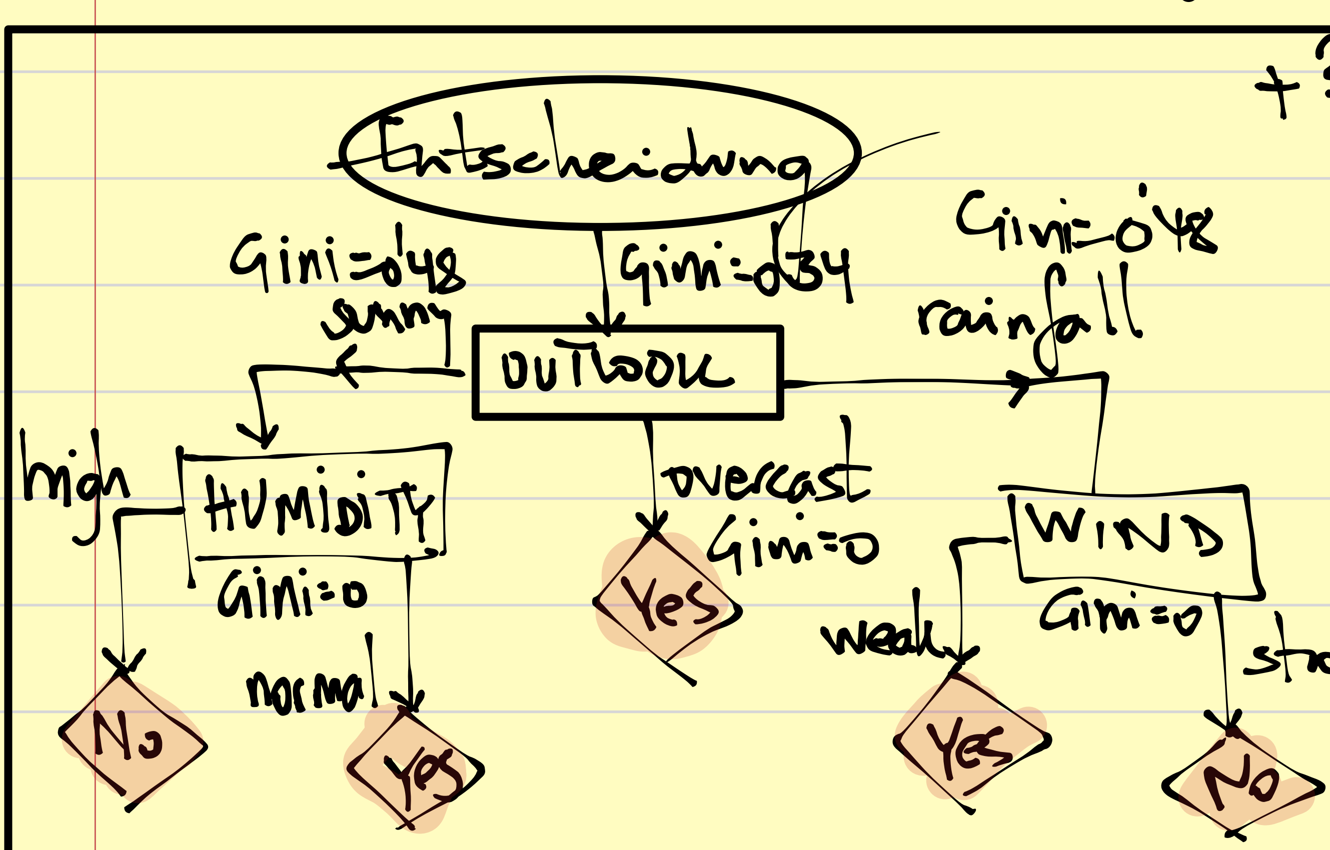
/5

$$\text{Gini}(\text{Outlook Sunny} + \text{Temp}) = \frac{2}{5} \cdot \left[1 - \left(\left(\frac{0}{2}\right)^2 + \left(\frac{2}{2}\right)^2\right)\right] + \frac{2}{5}\left[1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right)\right] + \frac{1}{5}\left[1 - \left(\left(\frac{1}{1}\right)^2 + \left(\frac{0}{1}\right)^2\right)\right] = 0'2$$

2. OUTLOOK SUNNY + HUM

	Yes	No	#
high	0	3	3
normal	2	0	2

$$\text{Gini}(\text{Outlook SUNNY} + \text{HUM}) = \frac{3}{5}\left(1 - \left(\left(\frac{0}{3}\right)^2 + \left(\frac{3}{3}\right)^2\right)\right) + \frac{2}{5}\left(1 - \left(\left(\frac{2}{2}\right)^2 + \left(\frac{0}{2}\right)^2\right)\right) = 0$$



OUTLOOK Rainfall + WIND	Yes	No	#
weak	3	0	3
strong	0	2	2
			/5

Gini(Outlook Rainfall + Wind) = 0

Exercise #2.

SEX: · Yes · No

	Kitchen clean	Groceries	Stress @ Work	Time Together	D
1.	dirty	big	strong	long	Y
2.	very dirty	small	mild	short	N
3.	dirty	small	mild	short	Y
4.	clean	big	weak	long	Y
5.	very clean	small	strong	long	N
6.	dirty	small	weak	short	N
7.	dirty	big	strong	long	Y
8.	clean	small	weak	short	N

Schritt 1. 1. KNOTE.

$$\text{Gini}(\text{KITCHEN}) = \underbrace{\frac{4}{8}\left[1-\left(\left(\frac{3}{4}\right)^2+\left(\frac{1}{4}\right)^2\right)\right]}_{\text{dirty}} + \underbrace{\frac{1}{8}\left[1-\left(\frac{1}{1}\right)^2\right]}_{\text{very dirty}} +$$

$$\underbrace{\frac{2}{8}\left[1-\left(\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2\right)\right]}_{\text{clean}} + \underbrace{\frac{1}{8}\left[1-\left(\frac{1}{1}\right)^2\right]}_{\text{very clean}} = 0'3125$$

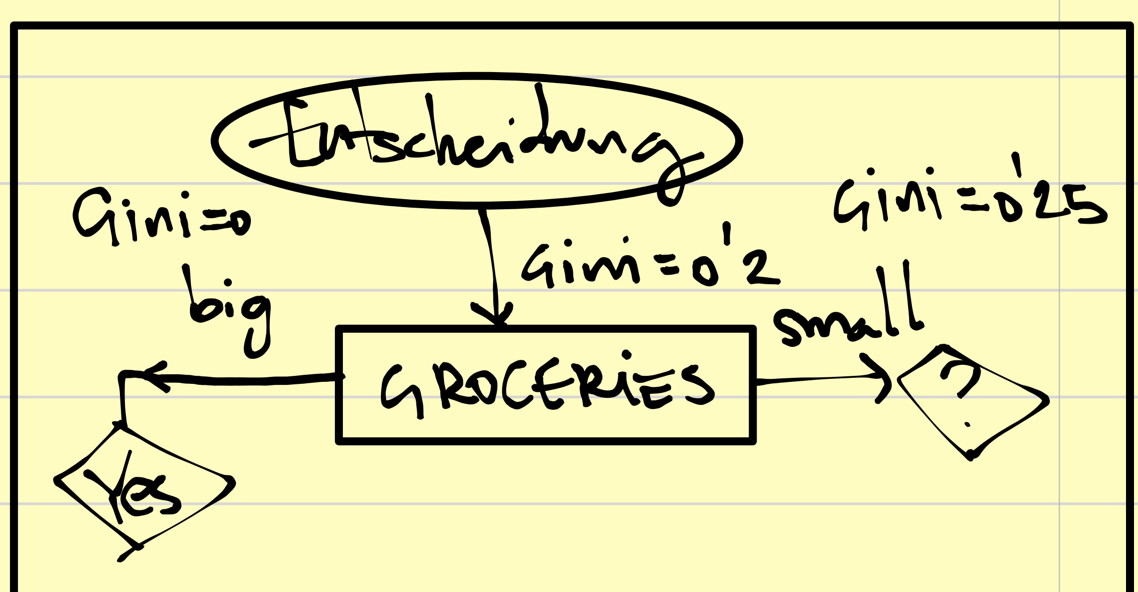
$$\text{Gini}(\text{GROCERIES}) = \overset{\text{big}}{\frac{3}{8}}\left[1-\left(\frac{3}{3}\right)^2\right] + \overset{\text{small}}{\frac{5}{8}}\left[1-\left(\left(\frac{4}{5}\right)^2+\left(\frac{1}{5}\right)^2\right)\right] = 0'2$$

mild

$$\text{Gini}(\text{STRESS}) = \overset{\text{strong}}{\frac{3}{8}}\left[1-\left(\left(\frac{2}{3}\right)^2+\left(\frac{1}{3}\right)^2\right)\right] + \frac{2}{8}\cdot\left[1-\left(\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2\right)\right] + \underset{\text{weak}}{\frac{3}{8}}\left[1-\left(\left(\frac{2}{3}\right)^2+\left(\frac{1}{3}\right)^2\right)\right] = 0'33$$

$$\text{Gini}(\text{TimeTogether}) = \underset{\text{long}}{\frac{4}{8}}\left[1-\left(\left(\frac{3}{4}\right)^2+\left(\frac{1}{4}\right)^2\right)\right] + \overset{\text{short}}{\frac{4}{8}}\left[1-\left(\left(\frac{3}{4}\right)^2+\left(\frac{1}{4}\right)^2\right)\right] = 0'375$$

	Gini
KITCHEN	0'3125
GROC.	0'2
STRESS	0'33
TIMETOG.	0'375



Schritt 2. Aus dem 1. Knote, wiederholen ...

Groc. small + stress	Y	N	#
weak	0	2	2
mild	1	1	2
strong	0	1	1 / 5

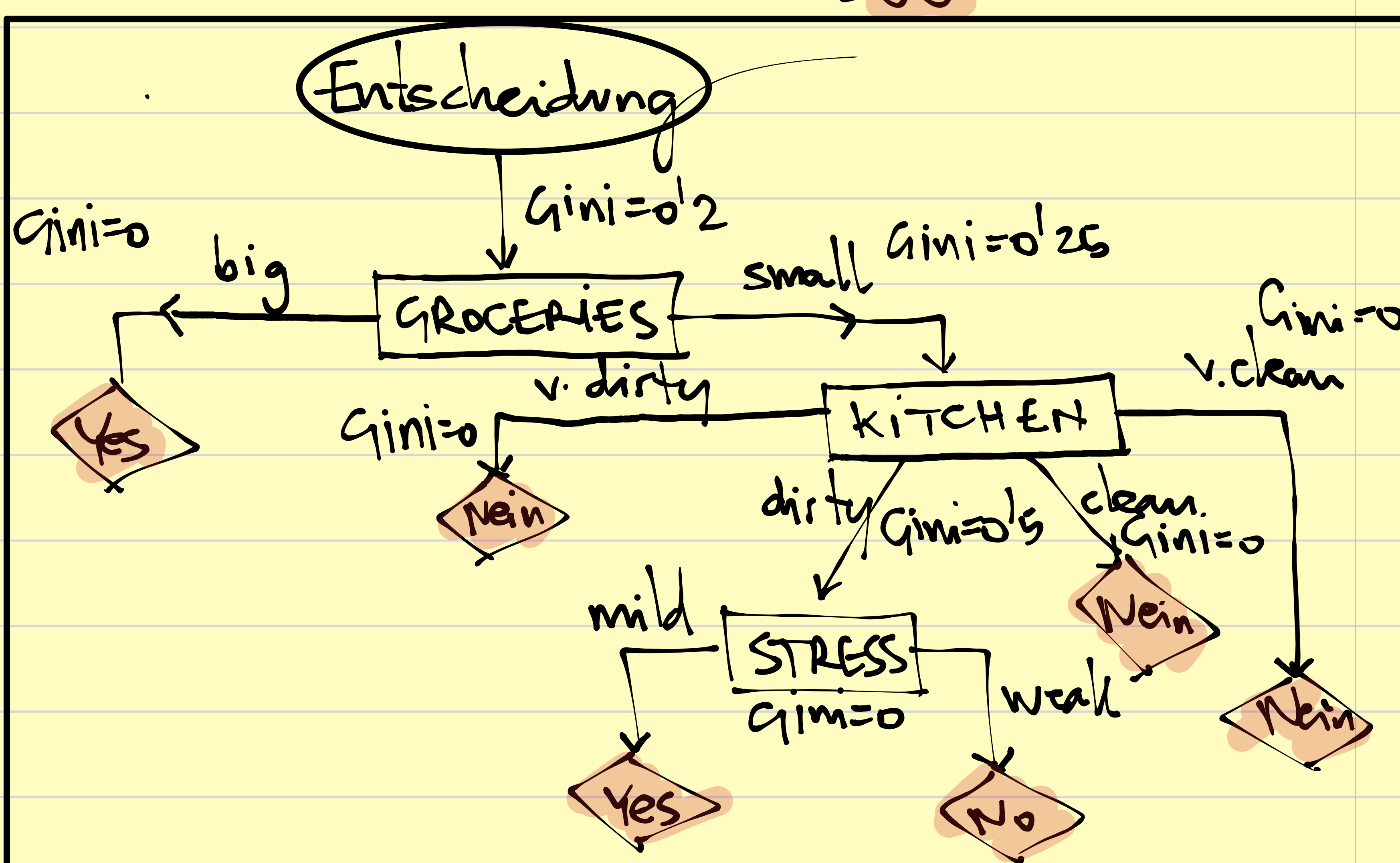
$$\text{Gini}(\text{groc.small} + \text{stress}) = \frac{2}{5}\cdot 0'5 = 0'2$$

Groc. small + Kitchen	Y	N	#
dirty	1	1	2
v dirty	0	1	1
clean	0	1	1
v clean	0	1	1 /5

$$\text{Gini}(\text{Groc.small+kitchen}) = \frac{2}{5} \cdot 0'5 = 0'2$$

Groc. small + Time	Y	N	#
short	1	3	4
long	0	1	1 /5

$$\text{Gini}(\text{Groc.small+Time}) = \frac{4}{5} \cdot \left[1 - \left(\left(\frac{1}{5}\right)^2 + \left(\frac{3}{5}\right)^2\right)\right] + 0 = = 0'3$$



HINWEIS FÜR DIE PRÜFUNG. Parametrisierung

MAT. NR.

2	3	7	4	5	6
a	b	c	d	e	f

$$\text{——}\ \underbrace{(2a+3)}_{\underbrace{2\cdot 2+3}_{7}}\ \text{——}\ \underbrace{(7c+d-f)}_{47}\ \text{——}$$